Lab 3 – User Manual

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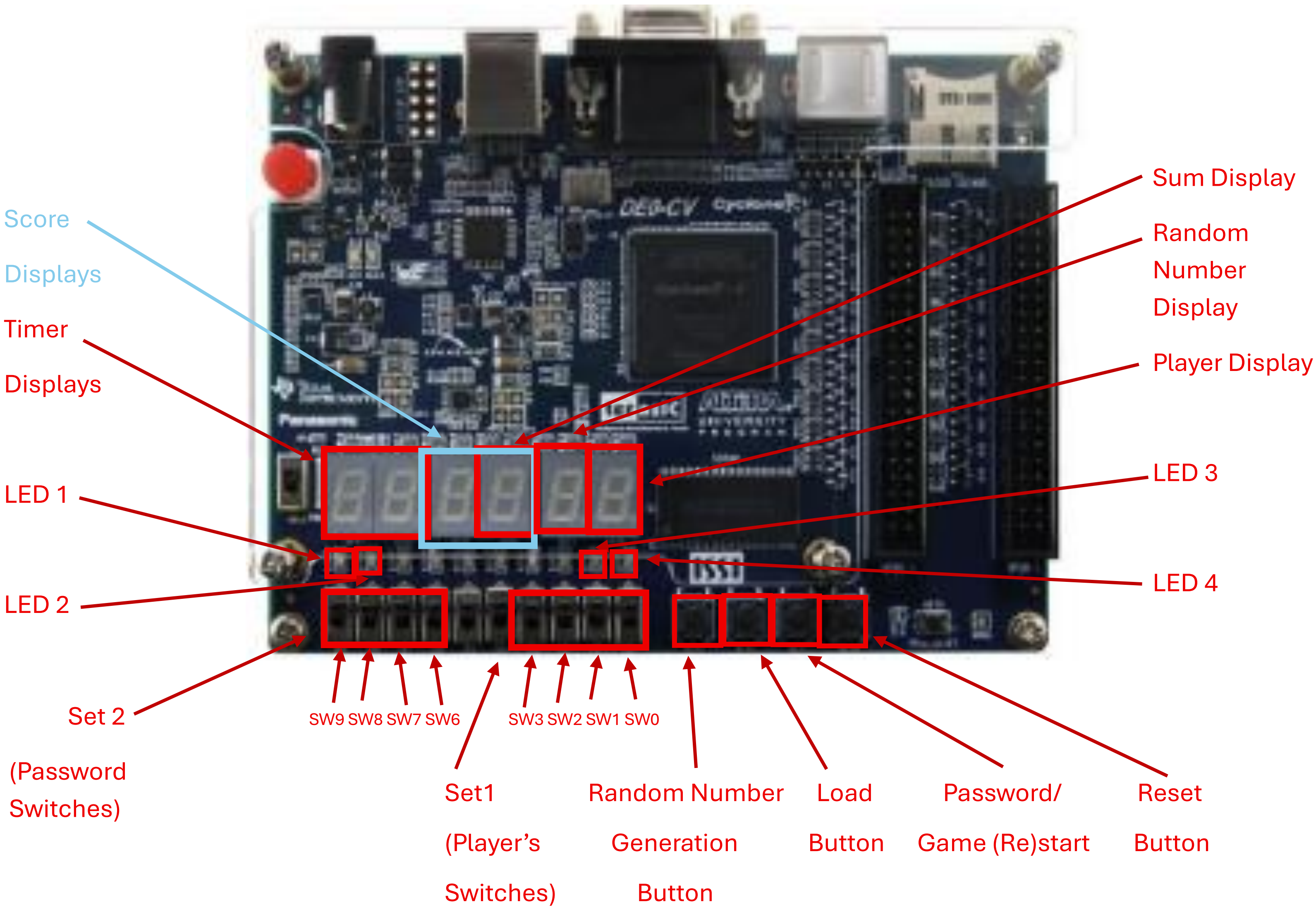
This system consists of a game of targeting a random number for the user on an FPGA board under a timer of 99 seconds. To start the game, the player need to authenticate themselves by entering the password, which is the last four digits of the creator's Peoplesoft ID, using the set of the four leftmost switches know as Set 2 (Figure 1). Each digit is set by toggling the four switches of Set 2 to high and low combinations so it matches the binary form of the corresponding digit. To enter that digit, push the third to left button on the right side of the board, which is a multifunctional button that initially works for password entering (Figure 1). This is done four times for each digit. If the authentication is successful, LED 4 (the rightmost light above the switches) will be turned on and the game can start, otherwise, LED 3 (the second rightmost) will remain on indicating that the players are still logged out and need to try to authenticate again.

If the user is successfully logged in, the timer (displayed by the two leftmost displays) will output the total amount of time they have to for the game (99 Seconds). It will remain displaying 99 until the user presses the multifunction button again (3^{rd} button from left on FPGA that works for game start in this moment) to start the game.

The game is played in rounds until the timer becomes 0. On each round, the player will press the Random Number Generation Button, which is the leftmost button on the FPGA). It will display a random hexadecimal number on the second rightmost display that varies depending on how long the user presses the button. Then the player will enter a number they believe that if summed to the random number generated by the Random Number Generation Button, the sum will result in 15. The player enters this number in its binary form by toggling the four rightmost switches on Set 1 (Figure 1). Then to enter that number, the player will push the Load Button, which is the second to left button at the right side of the board, (Figure 1). The resulting sum is displayed in hexadecimal form on the third rightmost display of the board. Additionally there are two LED lights above the switches that indicate whether the number displayed on the third display from right (Sum Display) is f_{16} or not. If it is f_{16}, LED 1 (the leftmost) will be on, if not, LED 2 (the second leftmost) will be on instead. The player can repeat the process in each round while the timer displayed on the two leftmost displays decreases by 1 each second. The system will keep registered how many times the player successfully matched the number for the sum to equal f_{16} (LED 1 on). When the timer displays 00, the game is over and the total of correct matches is displayed on the two displays at the center of the board. The user will not be able to enter another random generated number or a number from the switches on Set 1.

If the player wants to restart another game, they can press the multifunction button again (3^{rd} button from left on FPGA that works for game restart in this moment). It will set the timer back to 99. Then press the same button again to make the timer run so the player can restart playing.

To reset the system and log out of the game, press the reset button, which is the rightmost button on the board (figure 1), and LED 3 will be turned on indicating the users are logged out and LED 4 will be turned off.



Set 1 defines the switches used by Player 1 during the game:

SW3 – bit 3

SW2 – bit 2

SW1 – bit 1

SW0 – bit 0

Set 2 defines the switches used for authentication (log in):

SW9 – bit 3

SW8 – bit 2

SW7 – bit 1

SW6 – bit 0

Random Number Generation Button defines Player's button to generate a random number in the display for match.

Load Button defines Player's load button to register their input number in the display to match the random generated number.

Password/ (Re)start game button is pressed to enter a digit from the password during log in and to start and restart the game

Reset button resets the system and logs the user out.

LED 1 is the light when the sum equals 15.

LED 2 is the light when the sum does not equal 15.

LED 3 is the logged out indicator.

LED 4 is the logged in indicator.

How to log in (authenticate):

- Ensure LED 3 is on (logged out).
- Set SW9-SW6 to the corresponding binary of owner 's Peoplesoft ID's fourth to last digit.
- Press the Password Button (third button from left)

- Repeat the last two steps with the three last digits of owner 's Peoplesoft ID.
- If the entered digits correctly matches owner's last four digits of Peoplesoft ID, LED 4 will be turned on indicating players are logged in. Otherwise, LED 3 will stay on indicating players are logged out.

How to start the game:

- Ensure LED 4 is on (logged in).
- Ensure timer displays 99.
- Press the Start Button (third button from left)

Example of a failed round (Not an "F" sum):

- Player presses the Random Number Generation Button and the number 3_{16} on the Random Number display.
- Player sets the number 5_{16} on SW3-SW0 by toggling, from left to right, the first switch to low, the second switch to high, the third switch to low and the fourth switch to high. ($0101_2 = 5_{16}$)
- Player pushes Load Button.
- Player's number appears on Display.
- The number 8_{16} appears on Sum Display because $3_{16} + 5_{16} = 8_{16}$
- LED 2 is on because the sum is not equal to f_{16}
- Player loses the round for being unable to sum the random generated number to the value of f_{16}

Example of a successful round:

- Player presses the Random Number Generation Button and the number d_{16} on the Random Number display.
- Player sets the number 2_{16} on SW3-SW0 by toggling, from left to right, the first switch to low, the second switch to low, the third switch to high and the fourth switch to low. ($0010_2 = 2_{16}$)
- Player pushes Load Button.
- Player's number appears on Display.
- The number f_{16} appears on Sum Display (uppercase in this case) because $d_{16} + 2_{16} = f_{16} = 15_{10}$
- LED 1 is on because the sum is equal to f_{16}
- Player wins the round for being able to sum the random generated number to the value of f_{16} and one more point is added to the total score.

- The game continues as many rounds as the player can go while timer is not 00. When timer reaches 00, the total score is displayed on the Score Displays and the player cannot try another round.

How to start the game:

- Ensure LED 4 is on (logged in).
- Press the Restart Button (third button from left)
- Ensure timer displays 99.
- Press the Start Button (third button from left)

How to log out:

- Press the Reset button.
- LED 3 will be turned on and LED 4 will be turned off, indicating the players are logged out.

How to reset the system:

- Press the reset button.